SHORIFUL ISLAM

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ACADEMIC CREDENTIALS

Bangladesh Agricultural University (BAU) **Mymensingh, Bangladesh**
MS in Farm Structure and Environmental Engineering Oct 2024 – March 2026
- CGPA: **3.864/4.00**
- Dept. Rank: 1st out of 10 students

Bangladesh Agricultural University (BAU) **Mymensingh, Bangladesh**
Bachelor of Science (B.Sc.) with Honors in Agricultural Engineering Jan 2018 – July 2024
- CGPA: **3.639/4.00**
- Faculty Rank: 4th out of 89 Students

Relevant Coursework

Concrete Structure Design, Computer Aided Design, Foundation Engineering, Farm Building Design, Silo and Storage Structure, Environmental Engineering, Strength of Materials, Soil and Water Conservation Engineering, Fluid Mechanics, Irrigation and Drainage Engineering, Hydraulics, Groundwater Engineering, Solid Waste Management

RESEARCH INTEREST

- Neural Networks
- Machine Learning for Materials Optimization
- Predictive modeling using ML algorithms
- Concrete Structure Design
- Soil stabilization using bio-based materials
- Solid Waste Management

PUBLICATIONS

[1] **Islam, S.***, Kubra, M.K., Abedin, M.Z., Ashraf, M.A., & Hossain, M.B. (2026). Experiment-driven machine learning framework for compressive strength prediction of locally sourced fiber-reinforced concrete. Discover Civil Engineering, 3(1), 193. https://doi.org/10.1007/s44290-026-00583-y

[2] **Islam, S.**, Islam, R., Turag, K. A., Noyon, Md. J. H., Rana, M. M., Islam, R., Shariot-Ullah, M., & Amin, M. G. M. (2026). Effects of evapotranspiration-based supplemental irrigation intervals on Aman rice yield and groundwater recharge potential. *Paddy and Water Environment*. https://doi.org/10.1007/s10333-026-01079-2

[3] Abedin, M. Z., **Islam, S*.**, Rahman, M. Z., Ashraf, M. A., & Hossain, M. B. (2026). Predicting the cone penetration resistance of unsaturated agricultural soils by neural network. *Discover Soil.*, 3(1). https://doi.org/10.1007/s44378-026-00171-7

[4] Ridhy, N.A, **Islam S.***, Abedin, M.Z. (2025). Design and development of a low-cost water purifier using terracotta filter disc. *Bangladesh Rural Development Studies*, 28(2), 61-71

[5] **Islam, S*.**, Abedin, M. Z., & Ashraf, M. A. (2025). Predicting split-tensile strength of fiber reinforced concrete with lathe-waste steel scrap using artificial neural network. *Discover Concrete and Cement*, 1(1). https://doi.org/10.1007/s44416-025-00014-8

[6] Abedin, M.Z., Banarjee, A., & **Islam S.*** (2025). Coir-fiber reinforced soil-cement block as walling material for rural houses. *GSC Advanced Research and Reviews*, 23(1), 151–159. https://doi.org/10.30574/gscarr.2025.23.1.0098

FELLOWSHIP

National Science & Technology (NST) Fellowship MS/MSc (Food & Agricultural Science) 2025-2026
Ministry of Science and Technology, Government of Bangladesh;
Awarded for research project title:
"Development of FRC Paving Blocks Using Lathe-Waste Steel Fibers: An ANN-Based Mix Design"

CURRENT RESEARCH

- Development of fiber reinforced concrete paving block using lathe-waste steel scrap for sustainable use in walkway construction [BAURES Project; *No. 2025/28/BAU*]
- Data-Driven Prediction of Concrete Mix Proportions Using Machine Learning Framework
- Prediction of Compressive Strength of FRC with Lathe-waste Fiber Using Artificial Neural Networks

RESEARCH EXPERIENCES

MS Research Fellow (Research Assistant) Jan 2025 – Present

Concrete and Material Testing Laboratory, BAU

- Working on funded project: *Development of Fiber Reinforced Concrete Paving Block using Lathe-Waste Steel Scrap*
- Developed Machine Learning models for material prediction
- Conducted experimental testing of concrete and materials
- Performed data analysis using MATLAB/Python
- Prepared research manuscripts and technical reports

Research Assistant Jan 2022 – Dec 2023

Concrete and Material Testing Laboratory, BAU

- **Title:** Effect of Fiber Content on Strength Properties of fixed Mix Concrete Reinforced with Lathe-waste Steel Scrap
- Developed and compared ANN, SVM, GPR and tree-based models for strength prediction
- Published findings as *first author* in Springer Nature's Discover Concrete and Cement (DOI: 10.1007/s44416-025-00014-8)

CONFERENCES

[1] Afrin, F., **Islam, S.***, & Amin, M. G. M. (2026). *Leveraging ridge–furrow transplanting for enhancing rice water productivity in wet and dry seasons*. *In Proc. II International Scientific and Practical Conference on* "Rational Use of Natural Resources in Agriculture", Turkmen Agricultural Institute, Ashgabat, Turkmenistan, 22 April 2026.

[2] **Islam, S.***, Abedin, M.Z. & Ashraf, M.A. (2025). Predicting split-tensile strength of fiber reinforced concrete with lathe-waste steel scrap using artificial neural network. *In Proc. 1st International Conference on Agricultural Machinery and Bioresource Engineering (ICAMBE)*, Bangladesh Agricultural University, Mymensingh-2202, Bangladesh, 12-13 Feb (*Presenting author)

TECHNICAL SKILLS

- **Deep Learning & Machine Learning:** Predictive modeling using Artificial Neural Networks (ANN) and Machine Learning algorithms; TensorFlow.
- **Data Analysis & Visualization:** SPSS, Statistical analysis, data visualization, advanced MS Excel
- **Programming:** MATLAB, Python, C, C++, R
- **Design & Drafting:** AutoCAD, Adobe Photoshop, PowerPoint
- **Academic Tools:** Zotero, Mendeley Reference Manager, Microsoft Office Suite.

SCHOLARSHIPS & AWARDS

- **University Technical Stipend** (2019 – 2023): For having GPA above 3.50 out of 4.00 in the semester result.
- **Education Board Scholarship** (2013-2018): Provided by Board of Intermediate and Secondary Education, Jashore for extraordinary academic excellence in all of Junior School Certificate (JSC), Secondary School Certificate (SSC) and Higher Secondary School Certificate (HSC) Board Examinations.
- **Dutch-Bangla Bank Foundation Scholarship (**HSC 2017, SL# 1204): The Dutch-Bangla Bank Foundation (DBBF) Scholarship is a prestigious scholarship program in Bangladesh, provided by Dutch-Bangla Bank Limited (DBBL).

- **BRAC Medhabikash Scholarship** (11th Batch 2016, ID: 3902010211004137): BRAC providing this scholarship to promote students to get higher education.
- **Dutch-Bangla Bank Foundation Scholarship (**SSC 2015, SL# 1641): The Dutch-Bangla Bank Foundation (DBBF) Scholarship is a prestigious scholarship program in Bangladesh, provided by Dutch-Bangla Bank Limited (DBBL).

TRAININGS

Data Analysis: R & SPSS — 12 – 23 Oct, 2025
Conducted by:
Training Institute of Bangladesh Agricultural University

Basics of MS Office — 24 Aug – 4 Sept, 2025
Conducted by:
Training Institute of Bangladesh Agricultural University

Research Assistant — Jan 2022 – Dec 2023
Concrete and Material Testing Laboratory, BAU, Mymensingh
Appointment: 20 hours/Week
Major Responsibilities:

- Conducted advanced concrete testing and laboratory procedures
- Trained and supervised new lab members on experimental techniques
- Designed research methodology and experimental protocols
- Performed data analysis and drafted research manuscripts

Extension Field Trip — 12 – 17 March, 2023
Organized By: Dept. of Agriculture Extension Education, BAU
Duration: 5 Days

BRAC Development Program Training — 16 - 30 March, 2016
Information and Communication Technology & English Proficiency
Duration: 14 Days

EXTRA-CURRICULAR ACTIVITIES [Selected]

Former Membership Officer — March, 2021 – April 2023
IAAS Bangladesh, BAU

Former Writing & Publication Editor — June, 2023 – August 2024
Kushtia Zilla Somiti, Bangladesh Agricultural University, Mymensingh – 2202

Member — Since 2021
Engineers Foundation, Daulatpur, Bangladesh

PERSONAL INFORMATION

Date of Birth	September 18, 1999
Nationality	Bangladeshi
Mailing Address	440/D, Fazlul Haque Hall, Bangladesh Agricultural University, Mymensingh – 2202, Bangladesh

REFEREES

Dr. M.G. Mostofa Amin
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